

Business Sales Performance Analytics

Future Interns — Data Science & Analytics | Task 1 (FUTURE_DS_01)

Dataset: Sample Superstore | Tools: Python, Pandas, Matplotlib, Seaborn

Executive KPI Summary

Metric	Value	Metric	Value
Total Revenue	\$2,296,196	Total Profit	\$286,241
Total Orders	9,977	Avg Discount	15.6%
Profit Margin	12.47%	Top Category	Technology
Top Region	West	Top Segment	Consumer

Fig 1 — KPI Dashboard



Fig 2 — Category & Sub-Category Analysis

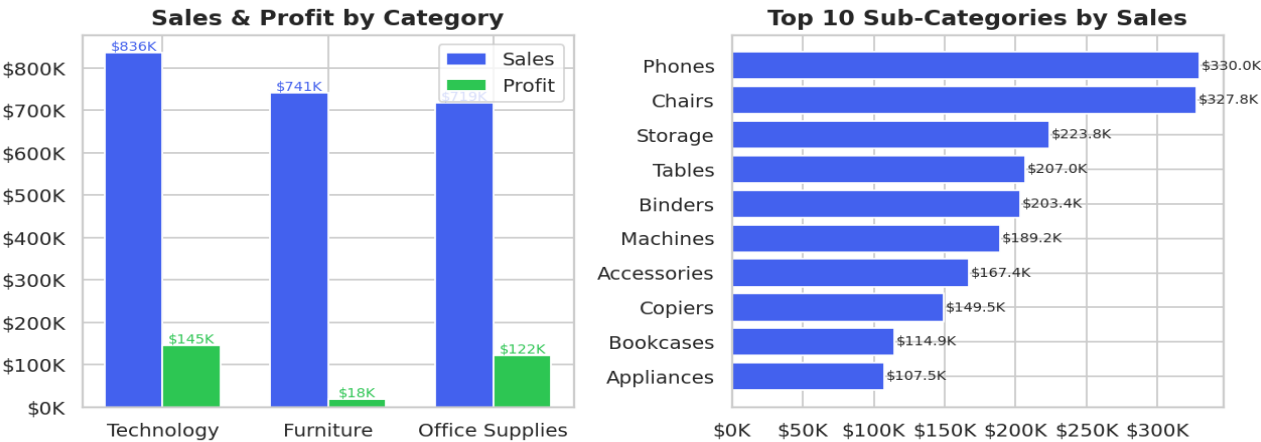


Fig 3 — Regional Performance Breakdown

Regional Performance Breakdown

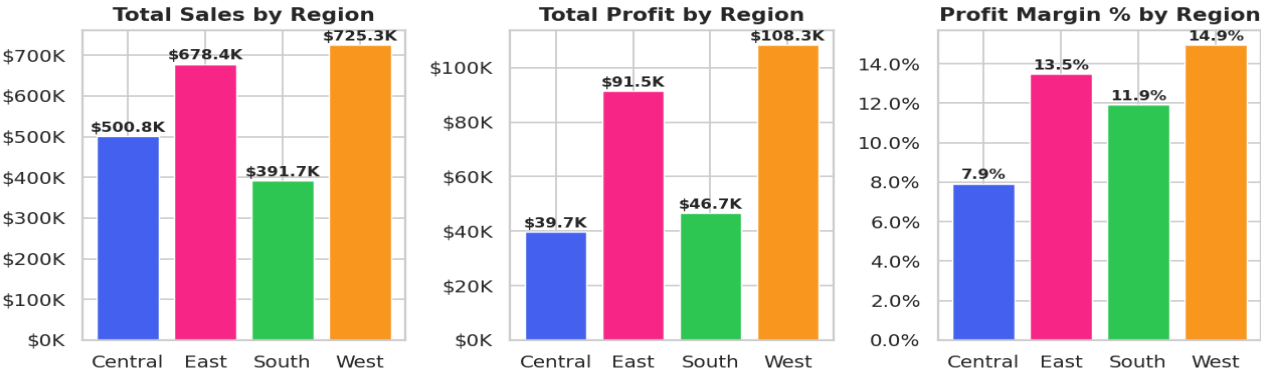


Fig 4 — Discount vs Profit Analysis

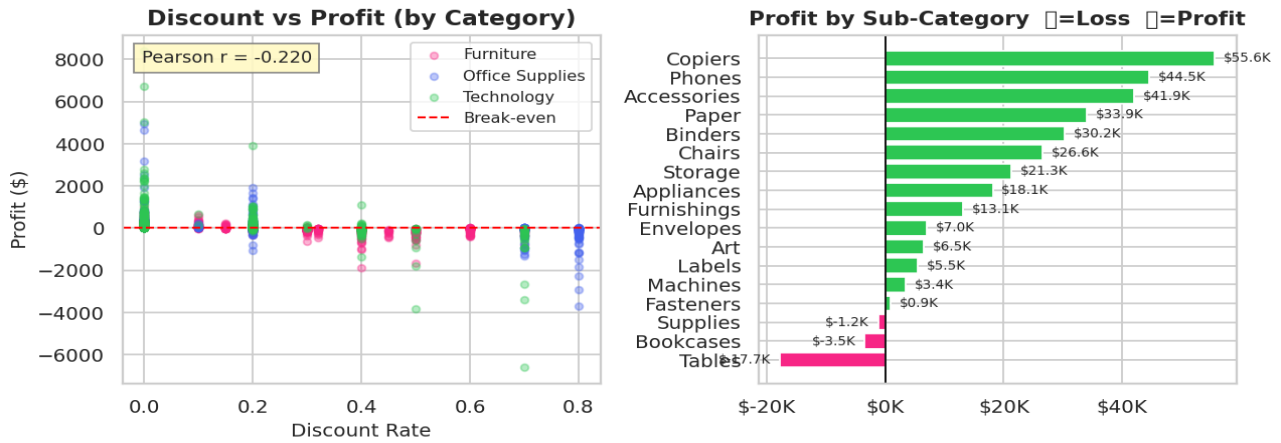


Fig 5 — Segment & Top States Analysis

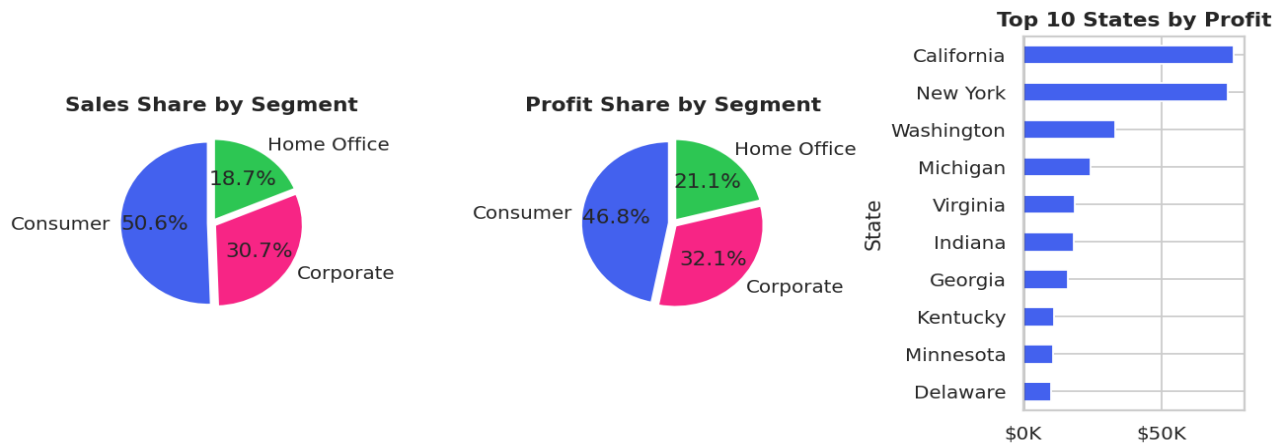
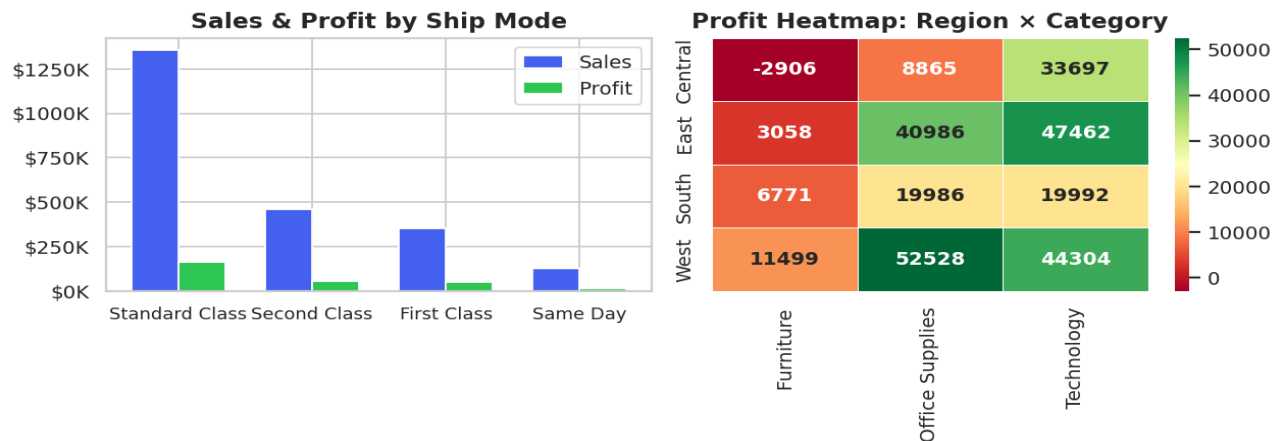


Fig 6 — Ship Mode & Region×Category Heatmap



Key Insights

- **Revenue Leader:** Technology is the top revenue and profit-generating category. Office Supplies follows in volume but with thinner margins.
- **Regional Champion:** The West region leads in both sales and profit. The Central region underperforms despite decent order volume.
- **Discount Impact:** Discount rate has a negative correlation with profit ($r = -0.220$). Orders above 40% discount almost always result in losses.
- **Loss-Making Products:** Sub-categories running at a loss: Tables, Bookcases, Supplies. These require an urgent pricing review.

■ **Best Segment:** Consumer segment generates the most revenue, but Consumer shows the strongest profit per order — ideal for targeted upselling.

■ **Shipping Efficiency:** Standard Class is the most used mode and generates the highest revenue. Same Day shipping has the lowest volume.

Actionable Recommendations

1. Cap Discounts at 20%: Implement a discount ceiling across all categories. Discounts above 40% consistently erode profitability.

2. Review Loss-Making SKUs: Conduct a pricing audit on Tables, Bookcases, Supplies. Consider bundling or discontinuing low-margin items.

3. Invest in West & East Regions: These two regions deliver the highest ROI. Allocate more marketing budget and inventory here.

4. Upsell Technology to Corporate Clients: Technology + Corporate is the highest-margin combination. Build targeted B2B campaigns around this segment.

5. Fix Central Region Performance: Central has high order counts but low margins — likely due to excessive discounting. Apply stricter discount controls here.